

WC Fantasy League 2026

Complete System Flow & Scoring Guide

Pick your Playing XI from two teams

Choose a Captain for 2x points

Admin updates scores with one click

Points auto-calculated & leaderboard updates

Built with Spring Boot + Angular + Material UI

Auto-tested with Playwright (video recordings included)

How It Works

Step 1: User Logs In

User enters a username on the login page. The system creates an account (or logs in if existing) and issues a JWT token for authentication.

Step 2: Browse Matches

The Matches page shows all upcoming FIFA World Cup 2026 group stage matches with team names, flags, venue, and date. Matches before kickoff show "Open" status with a "Pick Your XI" button.

Step 3: Pick Your Playing XI

User clicks "Pick Your XI" for a match. They see all players from BOTH teams (Team A + Team B). They must:

- Select exactly 11 players from the combined pool
- Designate one player as Captain (earns 2x points)
- Save the squad before the match starts

Once the match kicks off, the squad is locked automatically.

KEY RULE: Squad must be saved BEFORE match start time. After kickoff, no changes allowed!

Step 4: Admin Updates Scores (One Click)

After the match is completed, an admin goes to the Admin Panel (/admin) and clicks "Auto-Update Scores" for that match. This single click:

1. Fetches/generates player performance stats (goals, assists, cards, etc.)
2. Saves stats for every player in both teams
3. Marks the match as COMPLETED with final score
4. Triggers automatic point calculation for ALL user squads

Scoring System

Each player in your squad earns points based on their real match performance:

Action	Points	Applies To
Goal Scored	+6	All
Assist	+4	All
Clean Sheet	+4	GK, DEF
Man of the Match	+3	All
Appearance (played)	+1	All
Yellow Card	-1	All
Red Card	-3	All
Captain Bonus	x2	Captain Only

CAPTAIN BONUS: Your chosen captain gets ALL their points doubled!

Example: Captain scores a goal = $(6 + 1 \text{ appearance}) \times 2 = 14$ points

Example Calculation

Match: Brazil 2 - 0 Germany

User's Squad (11 players selected):

Player	Role	Performance	Points	With Captain
Vinicius Jr	FWD (C)	1 Goal + Appearance	7	14
Paqueta	MID	1 Goal + MOM + App.	10	10
Alisson	GK	Clean Sheet + App.	5	5
Marquinhos	DEF	CS + 1 Assist + App.	9	9
Militao	DEF	Clean Sheet + App.	5	5
Casemiro	MID	Appearance only	1	1
Raphinha	FWD	Appearance only	1	1
Neuer	GK	Appearance only	1	1
Kimmich	MID	App. + Yellow Card	0	0
Musiala	MID	Appearance only	1	1
Havertz	FWD	Appearance only	1	1

TOTAL POINTS EARNED: 48 pts

These 48 points are added to the user's total. The leaderboard ranks all users by cumulative points across all matches.

System Architecture

Tech Stack

Backend: Spring Boot 3.3 (Java 17) + Spring Data JPA + H2/PostgreSQL

Frontend: Angular 21 + Angular Material + RxJS

Auth: JWT (stateless)

Testing: Playwright (E2E with video recording)

API Endpoints

POST /api/auth/login	- Login/Register user
GET /api/matches	- List all matches
GET /api/players/team/{id}	- Players by team
POST /api/squads	- Save user squad (11 players)
GET /api/squads/{uid}/{mid}	- Get saved squad
GET /api/leaderboard	- Ranked users by points
POST /api/admin/update-scores/{id}	- ONE CLICK score update
GET /api/admin/match-stats/{id}	- View player stats

Auto-Update Flow (Behind the Scenes)

When admin clicks "Auto-Update Scores":

1. If FOOTBALL_API_KEY is set -> Fetches real stats from api-football.com
2. If no API key -> Generates realistic random stats (deterministic per match)
3. Saves MatchPlayerStats for every player in both teams
4. Updates Match status to COMPLETED with final score
5. Loops through ALL UserSquads for that match
6. Computes points per player using scoring rules
7. Doubles captain points
8. Updates each user's totalPoints
9. Returns result to frontend instantly

The entire flow completes in < 2 seconds with a single API call!